

780, Matlab or Python assignment 2025

Data Classification using machine learning with MATLAB or Python

The goal of this assignment is to make you write a MATLAB or Python code to build classification model of data using multivariate analysis and machine learning classification tools.

The deadline for handing in your submissions is 12 pm on Friday 23rd May 2025. It should be submitted to Canvas. You should also include a report answering the questions and a copy of the codes that you used. The codes should be written in Python or MATLAB (or other programming languages) with proper comment syntax that explains what your code is doing. You should send your code files in a format that I should be able to run. For example, comment clearly where the path should be changed for the code to run on my computer.

This assignment will make use of the following dataset: MIRFreshMeats from Shu, Hong-ping; Tang, Hong; Zhang, Haiqing; Zheng, Wenbin (2019), "Data for: Spectra Data Classification with Kernel Extreme Learning ", Mendeley Data, v1 <http://dx.doi.org/10.17632/frv2yd9rg.1>. The link above includes a description of the file that you will use.

As a side note, it shows the importance of sharing data for the benefit of the research community.

This dataset is described in "Mid-infrared spectroscopy and authenticity problems in selected meats: a feasibility study" Al-Jowder O, Kemsley E K, Wilson R. H.(1997) Food Chemistry 59 195-20 that use MIR spectra to classify meat. Note that we only use the fresh meat data for this assignment not the entire dataset used in the paper.

Assignment

1. Perform a PCA analysis of the dataset as a function of meat types. The PCA analysis should show the loadings and the scree-plot to justify how many PCs you use to plot the PCS results. Discuss the difference you see between your results and Figure 2 of the paper.
Note: Make sure you normalise correctly your data. Check what the software functions are doing exactly.
2. Classify the data between a bird (chicken and turkey) and not a bird (pork) using k-NN. Use cross-validation and testing as mentioned in the lectures. Discuss how "k" was optimised.
3. Classify the data between a bird (chicken and turkey) and not a bird (pork) using SVM. Justify the number of PCAs you use. Use cross-validation and testing as mentioned in the lectures. Discuss the cost parameter that was optimised. Test two kernels and explain which one you would choose and why. We recommend you try Linear Kernel and Radial Basis Function (RBF) Kernel.
4. Discuss which method you found to be the most efficient and which one you would have chosen in this case and why.

NOTE: All the MATLAB or Python (or any other appropriate programming language) codes you write should be handed in with your assignment. The code should be standalone, clearly commented on, explained, and justified, so that we can run it on any computer. Include your family name in your filename.